

Resampling-Based Model Selection for PE SDFs

How to empirically measure the risk-adjusted performance of non-traded cash flows

July 7, 2026

1 Introduction

Which stochastic discount factor (SDF) model gives the most credible risk adjustment for private-equity (PE) cash flows? Before that question can even be posed to the data, a more basic one must be answered: *how should the pricing moments be constructed?* PE samples are small, lumpy, and organized by vintage year, and an SDF model can look good or bad depending on how funds are grouped into pricing moments. Existing approaches commit to one grouping—single funds [Ang et al., 2018] or vintage-year portfolios [Driessen et al., 2012]—and treat it as fixed. This paper shows that the grouping choice is not innocuous, and that it has an answer.

We derive the resampling design from three finite-sample requirements and arrive at *blocked vintage cross-validation with pooled moments*: funds are pooled into one pricing moment per vintage; candidate SDFs are estimated on the vintage moment vector with a contiguous block of vintages (plus a buffer) held out; the held-out vintages are scored with an exactly unbiased estimate of their squared population pricing error; and all admissible blocks are enumerated. The preferred model is the one with the lowest average held-out pricing loss. The three requirements, developed in Section 4.2, are worth stating up front because each eliminates a natural competitor design:

1. **Pool what is exchangeable.** Funds within a vintage are exchangeable, so the pooled vintage moment is sufficient for the vintage’s pricing error: any random within-vintage sub-grouping equals the pooled moment plus mean-zero partition noise, and conditioning on the pooled moment weakly reduces every convex risk (Rao–Blackwell)—a finite-sample statement requiring no asymptotics. Randomizing fund cells *inside* the estimation moments is therefore strictly dominated; it also creates a rank-deficiency trap, because stacked exchangeable fold moments share a single population moment and cannot identify a multi-parameter SDF (Appendix A).
2. **Hold out what identifies.** A holdout can discriminate between SDF specifications only if the held-out moments embody variation the estimation moments do not. The identifying variation is cross-vintage—each vintage lives through a different factor path—so the holdout must be at the vintage layer. Fund-level holdouts within the same vintages hold the factor history fixed: whatever a specification fits about the vintage moments, including spurious fit to the single realized factor path, transfers one-for-one to the holdout, and the expected score gap between models collapses to their in-sample gap as fund counts grow. Because adjacent vintages overlap in calendar time, the held-out vintages form a contiguous block separated from the estimation set by a buffer, in the spirit of *hv*-block cross-validation for dependent data [Racine, 2000]; and the block must be sizeable, because with twenty-odd vintages a leave-one-out design leaves all candidate models fitting nearly the same sample.
3. **Score without a noise floor.** The squared held-out vintage moment estimates the squared population pricing error *plus* a fund-sampling noise floor that depends on fund counts and on the model itself. A U-statistic over fund pairs removes the floor exactly, so scores are unbiased for the squared population error conditional on the vintage’s factor history and comparable across vintages, models, and designs.

In short: *pool the exchangeable, hold out the identifying variation, score without bias.* Design choices that can be derived, we derive—the resampling scheme itself. Design choices that cannot, we rank empirically

by held-out pricing loss: the SDF factor structure, the moment family, the treatment of young funds, the alpha definition.

Two design problems confront every PE SDF study, and we resolve both inside the framework.

Unseasoned vintages. Recent vintages carry valuable pricing information—they reflect current market conditions and extend the factor-path coverage of the panel—but their value is still locked in appraised net asset values (NAVs). The standard treatment discounts the latest NAV back to the fund’s inception date as a terminal cash flow. This concentrates a large share of each young fund’s discounted value at a single calendar date, the dataset endpoint, and it treats an appraisal as if it were a realized payoff. Researchers therefore face a dilemma: exclude recent vintages and waste data, or include them and accept fragile, NAV-driven estimates. We propose a *hybrid* estimator: seasoned vintages contribute realized-cash-flow NPV moments, while unseasoned vintages contribute normalized one-period Euler errors on quarterly NAV returns—short-horizon restrictions that never discount an appraisal over a long horizon. The hybrid is disciplined by an evaluation principle we state as a design rule: *the criterion used to compare models and designs never touches a NAV*. In the blocked scheme this rule is satisfied by construction, because unseasoned vintages enter estimation only and are never scored; out-of-time (rolling-origin) validation on subsequently realized cash flows makes the principle operational prospectively.

Alpha design. The baseline parameterizes abnormal performance as a cumulative, NPV-style alpha in the direct-alpha and generalized-PME tradition [Gredil et al., 2023, Korteweg and Nagel, 2016, 2025], not as a per-period intercept inside the one-period discount factor as in Driessen et al. [2012]. Besides its natural long-horizon interpretation, the cumulative design has a practical advantage: it does not suffer from the *exploding-alpha* degeneracy of the intercept parameterization, in which driving $\alpha \rightarrow \infty$ sends all discounted cash flows to zero, so that each fund’s pricing error collapses to its undiscounted time-zero flows and a ridge of spurious near-minima appears whenever those undiscounted flows are small. Under the baseline design with time-zero indexing, the undiscounted initial capital call anchors the objective away from zero, and the Euler block actively penalizes extreme alpha. We therefore treat exploding alpha as a solved design issue in the main text and relegate its full anatomy, together with identification-robust inference for alpha, to Appendix B.

The empirical strategy has two parts. A compact *semi-synthetic simulation* (Section 7) holds the real design fixed—historical factor paths, the actual vintage structure and fund counts of the estimation panel—and synthesizes only cash flows under a known SDF. It delivers the three quantities no real-data check can: selection frequencies (does the criterion find the true factor model?), small-sample bias of the estimated alpha and loadings, and the size of the corrected tests. It also verifies the derivation’s central prediction empirically: fund-level cross-validation fails to discriminate factor models, while blocked vintage cross-validation succeeds. The *empirical application* (Section 8) then estimates and ranks candidate SDF models on a buyout (BO) fund sample, with diagnostics that separate genuine pricing ability from design artifacts.

The paper makes four contributions.

1. **A resampling framework for PE SDFs, derived rather than asserted.** Blocked vintage cross-validation with pooled moments and unbiased U-statistic scoring, obtained from three finite-sample requirements; the block enumeration is exhaustive, so no Monte Carlo tuning parameter remains. Inference for score differences corrects the anti-conservative naive resampled *t*-statistic [Bengio and Grandvalet, 2004, Nadeau and Bengio, 2003] and reports a model confidence set [Hansen et al., 2011].
2. **An identification analysis of resampled pricing moments.** Exchangeable fund-level folds are redundant moments in the precise GMM sense; the vintage cross-section carries the identifying rank. Pooling implements the fix by construction (Appendix A gives the formal treatment).

3. **A hybrid treatment of unseasoned vintages with NAV-free evaluation.** Realized cash flows are used wherever they exist; NAV information enters only through short-horizon Euler restrictions on young funds; and no NAV ever enters a comparison score.
4. **Simulation and empirical evidence:** semi-synthetic verification of the framework’s selection and inference properties, and an empirical ranking of SDF models for buyout funds.

Related literature. The paper is closest in spirit to Korteweg and Nagel [2016] because the estimator works with aggregated pricing moments and an NPV-style abnormal-performance object, and to Driessen et al. [2012] because the moments price private cash flows directly; Ang et al. [2018] estimate PE returns from fund-level cash flows in a Bayesian time-series framework, and Kaplan and Schoar [2005] introduce the PME benchmark that underlies the ratio moment family. On the NAV side, Brown et al. [2023] nowcast fund NAVs and Ghysels et al. [2025] construct high-frequency PE return indices from NAV-based state-space models; our Euler block uses NAV returns only for young funds and only over one-quarter horizons. On the methodology side, we import results from the cross-validation literature: dependent-data blocking [Racine, 2000], the insight that small holdout fractions overfit in model selection [Shao, 1993, Arlot and Celisse, 2010], the impossibility of unbiased variance estimation for resampled scores [Bengio and Grandvalet, 2004] and its practical correction [Nadeau and Bengio, 2003], and the analysis of bagging [Bühlmann and Yu, 2002]. Relative to all of these, the new element is a resampling design derived for the PE moment structure—pooled vintage moments, blocked vintage holdout, unbiased scoring—rather than an off-the-shelf cross-validation bolted onto an existing estimator.

Section 2 gives a guiding example. Section 3 sets out the pricing moments. Section 4 develops the blocked vintage cross-validation scheme, its optimality argument, identification, and inference. Section 5 develops the hybrid treatment of unseasoned vintages and the NAV-free evaluation target. Section 6 explains the cumulative alpha design. Section 7 presents the semi-synthetic simulation design and Section 8 the buyout application. Section 9 concludes. Appendices collect the formal identification analysis, the exploding-alpha anatomy, alternative unseasoned-vintage treatments, the moment-family comparison, and a research agenda.

2 Guiding Example

Suppose a candidate SDF has three parameters: a cumulative alpha a and two factor loadings,

$$\theta = (a, \beta_1, \beta_2)'$$

The baseline one-period discount factor is the cumulative-alpha (direct-alpha-style) form

$$M_{t-1,t}(\theta) = \frac{e^{-a \Delta_t}}{1 + r_{f,t} + \beta_1 f_{1,t} + \beta_2 f_{2,t}}, \quad \Psi_{0,t}(\theta) = \prod_{s=1}^t M_{s-1,s}(\theta) = \frac{e^{-a \tau_t}}{B_{0,t}(\beta)},$$

where $r_{f,t}$ is the risk-free rate, $f_{1,t}$ and $f_{2,t}$ are excess returns on two traded factor portfolios, Δ_t is the period length, τ_t the elapsed time from the vintage inception date, and $B_{0,t}(\beta)$ the cumulative benchmark return implied by the loadings. Setting $a = 0$ recovers pure benchmark discounting; a is a continuously compounded abnormal return earned on top of the factor benchmark (Section 6).

The panel in Figure 1 has eight active vintages: 2005–2010 are seasoned ($\mathcal{V}^S$) and 2011–2012 are unseasoned ($\mathcal{V}^U$). Write $CF_{t,i} = D_{t,i} - X_{t,i}$ for fund i ’s net cash flow (distributions minus contributions) and K_i for its committed capital. Every vintage contributes *one* pooled pricing moment. For a seasoned vintage it is the

average relative NPV error over *all* of its n_v funds,

$$\epsilon_v(\theta) = \frac{1}{n_v} \sum_{i \in I_v} \frac{1}{K_i} \sum_{t=0}^{T_i} \Psi_{0,t}(\theta) CF_{t,i}, \quad v \in \mathcal{V}^S,$$

with the time-zero capital call entering undiscounted ($\Psi_{0,0} = 1$). For an unseasoned vintage it is a normalized one-period Euler error built from quarterly NAV returns using the same one-period SDF (Section 5); unseasoned vintages support estimation but are never scored.

The scheme holds out whole vintages in contiguous blocks. With block length $d = 2$ and a buffer of $g = 1$ vintage on each side, the block $\{2007, 2008\}$ shown in the figure implies buffer vintages $\{2006, 2009\}$ (excluded entirely) and the estimation set $\{2005, 2010, 2011, 2012\}$. The estimator minimizes the squared pooled vintage moments on the estimation set,

$$\hat{\theta}_b = \arg \min_{\theta \in \Theta} \frac{1}{|\mathcal{E}_b|} \sum_{v \in \mathcal{E}_b} \epsilon_v(\theta)^2,$$

which here stacks four moment conditions—two seasoned NPV moments and two unseasoned Euler moments—against three parameters; identification comes from the four vintages’ distinct factor paths (Section 4.4). The held-out block is then scored with an *unbiased* estimate of each held-out vintage’s squared population pricing error,

$$S_b(\hat{\theta}_b) = \frac{1}{2} [\widehat{U}_{2007}(\hat{\theta}_b) + \widehat{U}_{2008}(\hat{\theta}_b)],$$

where $\widehat{U}_v$ averages products of fund-pair pricing errors within the vintage and thereby strips out the fund-sampling noise floor (Section 4.3).

There is no Monte Carlo in this construction: the admissible blocks are simply the five contiguous windows $\{2005\text{--}06\}$, $\{2006\text{--}07\}$, $\{2007\text{--}08\}$, $\{2008\text{--}09\}$, $\{2009\text{--}10\}$, each is estimated and scored in turn, and models are ranked by the average score across the five blocks. In the empirical application the seasoned range is longer (around twenty vintages) and the baseline block holds out roughly one-fifth of it.

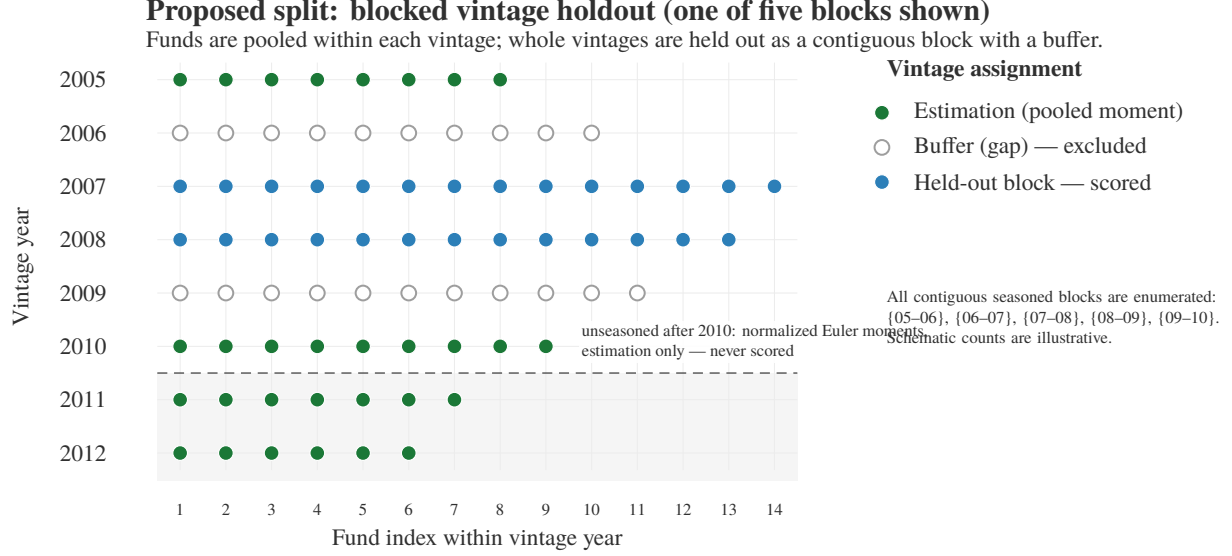


Figure 1: Blocked vintage holdout, showing one of the five admissible blocks. Funds are pooled within each vintage into one pricing moment. The held-out block {2007, 2008} (blue) is scored; the buffer vintages 2006 and 2009 (open circles) are excluded because their funds’ lives overlap the block almost completely; the remaining vintages (green) form the estimation set. Unseasoned vintages 2011–2012 (grey band) contribute normalized one-period Euler moments to estimation but are never scored, so every reported score is built from realized cash flows only.

3 Pricing Moments

3.1 Funds, vintages, and seasoning

Let $v = 1, \dots, V$ index active vintage years, ordered by year, and $\mathcal{I}_v$ the n_v funds of vintage v , each with committed capital K_i , strategy $s(i)$, and quarterly contributions $X_{t,i}$, distributions $D_{t,i}$, and reported NAVs. Let M denote a candidate SDF model with parameters $\theta \in \Theta_M \subset \mathbb{R}^{p(M)}$, and let η collect the remaining design choices (moment family, weighting, seasoning threshold, terminal-NAV convention).

A vintage v is *unseasoned* ($v \in \mathcal{V}^U$) if its aggregate residual NAV is large relative to the value it has already realized; otherwise it is *seasoned* ($v \in \mathcal{V}^S$). A convenient scale-free criterion is the unrealized-value fraction $\text{RVPI}/(\text{RVPI} + \text{DPI})$ exceeding a threshold δ ; scaling residual NAV by committed capital is an alternative, but that denominator can remain near or above one for strong funds long after they are effectively mature. In practice, $\delta \in [0.10, 0.20]$ is a sensible range. “Seasoned” therefore does not require residual NAV to be exactly zero; it means the remaining NAV is small enough that the vintage is treated as effectively mature.

How to handle the small residual NAV of a seasoned vintage is itself a design choice in η : exclude it, include it as a final distribution, or apply a de minimis adjustment. The important requirement is consistency: once a terminal-NAV convention is chosen, it must be applied identically in estimation, scoring, robustness checks, and common-score comparisons.

3.2 Fund-level pricing errors

For seasoned funds, the baseline contribution is the relative NPV error per dollar committed,

$$g_i^S(\theta) = \frac{1}{K_i} \sum_{t=0}^{T_i} \Psi_{0,t}(\theta) CF_{t,i}, \quad v(i) \in \mathcal{V}^S,$$

with the cumulative SDF $\Psi_{0,t}(\theta)$ of Section 2 and the time-zero call entering undiscounted. At the population parameter, $E[g_i^S(\theta_0)] = 0$: the risk-adjusted NPV of fund cash flows vanishes. For unseasoned funds, the baseline contribution is the normalized one-period Euler error $g_i^U(\theta)$ constructed in Section 5.2: a fund-average quarterly NAV-return pricing error rescaled by a deterministic duration factor into the same per-committed-dollar units. The empirical baseline is therefore *hybrid*: cumulative NPV for $\mathcal{V}^S$, normalized Euler for $\mathcal{V}^U$. Both blocks are relative pricing errors, and neither requires ratio denominators that can approach zero.

The NPV moment is the baseline moment family; the PME-ratio family of Kaplan and Schoar [2005] and Ang et al. [2018] serves as a robustness check, restricted to seasoned vintages because integrating young funds into a PME requires discounting a terminal NAV back to inception—exactly the mechanism the Euler block avoids. The two families estimate different population objects (the expectation of a ratio is not the ratio of expectations), so agreement of the preferred model across families indicates genuine fit while disagreement flags moment-induced selection. Appendix D gives the detailed comparison.

3.3 Pooled vintage moments and weights

The primitive estimation object is the *pooled vintage moment*: the average pricing error over all funds of a vintage,

$$\epsilon_v(\theta) = \frac{1}{n_v} \sum_{i \in \mathcal{I}_v} g_i(\theta), \quad g_i = g_i^S \text{ or } g_i^U,$$

collected into the vintage moment vector $\epsilon(\theta) = (\epsilon_v(\theta))_v$ with weights ω_v . The baseline sets ω_v equal across vintages, so each vintage contributes equally regardless of fund count or aggregate capital; value weighting is a design alternative in η . Under this two-layer normalization (per-committed-dollar at the fund level, equal weights at the vintage level), the moments are comparable across (i) vintages with different numbers of funds, (ii) vintages with different aggregate capital, and (iii) seasoned cash-flow errors and unseasoned normalized Euler errors.

Why the *pooled* average over all funds, rather than sub-portfolios? Because funds within a vintage are exchangeable, any random sub-grouping produces moments that equal the pooled moment plus mean-zero noise—randomization inside the estimation moments only injects variance and, if sub-groups are stacked as separate moment conditions, a rank-deficiency trap. Section 4.2 makes both points precise; the constructive consequence is that all resampling in this paper operates *across* vintages, never within them.

The scalar aggregate $\sum_v \omega_v \epsilon_v(\theta)$ is *not* the estimation objective: collapsing the vector to its average hides sign cancellation (overpricing one vintage offsetting underpricing another) and destroys the identifying rank of the moment system (Section 4.4). We retain it only as a reported diagnostic: a small vector loss with a small scalar aggregate indicates uniform pricing; a small scalar aggregate with a large vector loss indicates cancellation.

4 Blocked Vintage Cross-Validation

4.1 The scheme

Order the seasoned vintages by year. A *block* is a contiguous window $\mathcal{H}_b \subset \mathcal{V}^S$ of d vintages; its *buffer* $\mathcal{G}_b$ consists of the g vintages adjacent to the block on each side; the *estimation set* is everything else,

$$\mathcal{E}_b = \mathcal{V} \setminus (\mathcal{H}_b \cup \mathcal{G}_b),$$

which always includes the unseasoned vintages $\mathcal{V}^U$. The admissible blocks are the $B = |\mathcal{V}^S| - d + 1$ contiguous windows in the seasoned range; the scheme enumerates all of them. Blocks are drawn only from $\mathcal{V}^S$: unseasoned vintages support estimation but are never scored, so every reported score is built from realized cash flows only (Section 5.3).

The buffer exists because vintages are not calendar-separated: a 2007 fund’s life overlaps a 2008 fund’s life almost completely, so holding out 2008 while estimating on 2007 would leak most of the held-out information back into estimation. Removing the immediate neighbors implements the logic of *hv*-block cross-validation for dependent data [Racine, 2000] at the vintage level. Complete separation is unattainable—distant vintages still share some calendar quarters, since funds live ten to fifteen years—so the buffer reduces rather than eliminates overlap; the rolling-origin design of Section 5.3, which truncates the information set in calendar time, is the fully separated (but one-sided) extreme of the same family, with the terminal block as its holdout.

Two tuning constants remain, and both are pre-registered rather than optimized: the block length d and the buffer width g . The baseline sets $d \approx 0.2 |\mathcal{V}^S|$ and $g = 1$; Section 4.2 explains why d must be a sizeable fraction of the panel, and the simulation (Section 7) documents sensitivity to both.

4.2 Why this scheme: three finite-sample requirements

The scheme is not one design among many; it is the one that survives three requirements, each of which eliminates a natural competitor. All three arguments are finite-sample—none appeals to large- V asymptotics, which would be out of place in a panel of twenty-odd vintages.

R1: Pool what is exchangeable. Fix any θ and any vintage v . Conditional on the vintage’s factor history, the funds’ pricing errors $g_i(\theta)$ are exchangeable draws around the vintage population error $\bar{g}_v(\theta)$. Any moment built from a random sub-group of funds is therefore the pooled moment plus mean-zero partition noise, and by Rao–Blackwell (conditioning on the pooled moment and applying Jensen’s inequality) replacing sub-group moments by the pooled moment weakly reduces every convex criterion—strictly, unless the sub-grouping is trivial. Appendix A quantifies the injected noise in closed form and adds the identification half of the argument: sub-group moments stacked as separate GMM conditions are *exchangeable duplicates* sharing one population moment, so their Jacobian has rank one and cannot identify $p \geq 2$ parameters. Estimation must therefore use the pooled vintage moment vector: this both minimizes the attainable fund-noise floor and removes the redundant-moment trap by construction.

R2: Hold out what identifies. A holdout discriminates between two SDF specifications only if the held-out moments respond to the specifications differently than the estimation moments do—otherwise the score comparison merely replays the estimation comparison on noisier data. Under fund-level holdouts within the same vintages, the estimation and holdout moments share identical factor histories: their population values coincide vintage by vintage, so whatever a specification fits about the vintage moments—including spurious fit to the single realized factor draw—transfers one-for-one to the holdout. The expected score gap between two models then equals their in-sample population gap, and the only thing the holdout penalizes is estimation

noise from finite fund counts, a penalty that disappears exactly when the data improve. Discrimination requires holding out *vintages*, whose factor paths the estimation set has genuinely not used. Two small-sample corollaries follow. First, the block must be contiguous with a buffer, because adjacent vintages are near-duplicates in calendar time (R2 would otherwise be undone by leakage). Second, the block must be a sizeable fraction of the panel: with $|\mathcal{V}^S| \approx 20$, a leave-one-vintage-out design estimates every candidate on 19 of 20 vintages, so all candidates produce near-identical fits and near-identical scores—no discrimination. Holding out 20–25% forces genuine extrapolation across factor paths.¹

R3: Score without a noise floor. The squared held-out moment is a biased estimate of what model selection needs:

$$E[\epsilon_v(\theta)^2 \mid \text{factor path}] = \bar{g}_v(\theta)^2 + \frac{\sigma_v^2(\theta)}{n_v},$$

where $\sigma_v^2(\theta)$ is the cross-fund variance of $g_i(\theta)$ within the vintage. The floor $\sigma_v^2(\theta)/n_v$ depends on the fund count *and on the model*, so raw squared scores conflate mean pricing error with cross-fund dispersion and are not comparable across vintages, models, or designs with different cell sizes. The fix is exact and costless: the U-statistic over fund pairs,

$$\widehat{U}_v(\theta) = \frac{1}{n_v(n_v - 1)} \sum_{i \neq j} g_i(\theta) g_j(\theta) = \epsilon_v(\theta)^2 - \frac{\widehat{\sigma}_v^2(\theta)}{n_v}, \quad E[\widehat{U}_v(\theta) \mid \text{factor path}] = \bar{g}_v(\theta)^2,$$

is unbiased for the squared population pricing error in finite samples. Where fund-level splitting survives, this is it: $\widehat{U}_v$ is precisely the average product of pricing errors across independent fund pairs—the two-half-samples idea reduced to its unbiased core. Two remarks. First, $\widehat{U}_v$ can be negative in samples where the vintage is priced well; that is the price of unbiasedness and is harmless for averaging and for paired comparisons (we report raw ϵ_v^2 alongside as a bounded-below companion). Second, unbiasedness requires cross-fund independence conditional on the factor path; funds sharing a general partner violate it mildly, and the refinement is to restrict the U-statistic to cross-GP pairs.

The fate of fund-level folds. The within-vintage fund-partition design of Figure 2—randomly splitting each vintage’s funds into estimation and validation cells—is dominated on all three counts: it randomizes inside estimation moments (against R1), its holdout shares the factor history with estimation (against R2), and its cell-level scores carry the largest noise floors (against R3). It nonetheless retains three legitimate jobs, and we use it for all three: the fund-pair structure inside $\widehat{U}_v$; *stability diagnostics* (the dispersion of $\hat{\theta}$ under fund resampling measures sensitivity to sample composition); and cross-fitting of any tuned nuisance objects. What it cannot do is select models, and the simulation of Section 7 documents this failure directly.

4.3 Estimator, score, and enumeration

For block $b = 1, \dots, B$, the split-level estimator minimizes the weighted squared pooled vintage moments on the estimation set over the admissible parameter region,

$$\hat{\theta}_b(M, \eta) = \arg \min_{\theta \in \Theta_M} \sum_{v \in \mathcal{E}_b} \omega_v \epsilon_v(\theta)^2,$$

where Θ_M imposes economically admissible restrictions fixed before any score is computed: positivity of the one-period benchmark denominator over the observed factor support and a bounded box for the cumulative

¹The same conclusion is known asymptotically: cross-validation with a vanishing holdout fraction overfits in model selection, and consistency requires the holdout fraction to remain large [Shao, 1993, Arlot and Celisse, 2010]. We emphasize the small-sample version of the argument: with few effective observations, small holdouts simply cannot separate candidate fits.

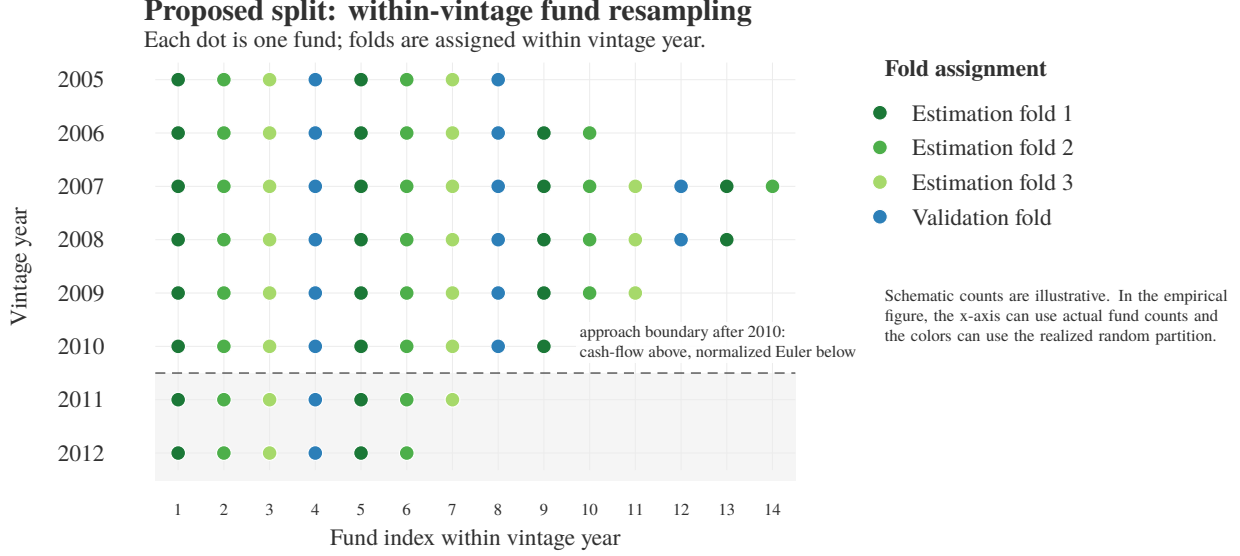


Figure 2: The dominated alternative: within-vintage fund-level folds. Each vintage’s funds are randomly split into estimation cells (greens) and a validation cell (blue). Estimation and validation cells share every vintage’s factor history, so the holdout cannot discriminate between factor specifications (requirement R2); random cells also inject partition noise into estimation (R1) and carry the largest fund-noise floors (R3). The design is retained only for stability diagnostics and nuisance cross-fitting.

alpha. (A bias-corrected variant that replaces ϵ_v^2 by $\hat{U}_v$ in the *objective* is a robustness option; with pooled moments the estimation-side floor σ_v^2/n_v is already the smallest attainable, so the baseline keeps the standard GMM objective.) The block score applies the unbiased statistic to the held-out vintages,

$$S_b(M, \eta) = \frac{1}{d} \sum_{v \in \mathcal{H}_b} \hat{U}_v(\hat{\theta}_b(M, \eta)), \quad \text{Score}(M, \eta) = \frac{1}{B} \sum_{b=1}^B S_b(M, \eta).$$

The preferred model-design pair is the one with the lowest average held-out loss: an unbiased estimate of the mean squared population pricing error of vintages the estimator has not seen, evaluated at parameters estimated without those vintages or their immediate neighbors. Because the blocks are enumerated exhaustively, there is no simulation noise in Score itself; the only randomness left in the pipeline is the bootstrap used for inference. Interior vintages appear in more blocks than edge vintages; we report per-vintage holdout coverage and, as a robustness check, a coverage-reweighted score.

All candidate models are estimated and scored on the identical block sequence, so comparisons are paired block by block. Model-specific diagnostics recorded per block are the estimated moment Jacobian’s condition number, boundary and non-convergence flags, and the seasoned/unseasoned split of estimation-side fit.

4.4 Identification: vintages carry the rank

It is tempting to think that more folds, cells, or splits mean more moment conditions. The opposite discipline applies: only *vintages* (or strategy–vintage cells) are distinct moments. Funds within a vintage are exchangeable, so any two sub-groups of the same vintage share one population moment and one Jacobian

row; stacking them adds no rank (Appendix A). Vintages are not exchangeable: each lives through a different factor path and cash-flow timing, so the sensitivities $\partial \bar{g}_v(\theta_0)/\partial \theta'$ differ across v . Local identification on estimation set $\mathcal{E}_b$ requires the cross-vintage Jacobian

$$J_b(\theta_0) = \frac{\partial}{\partial \theta'} (\bar{g}_v(\theta_0))_{v \in \mathcal{E}_b} \in \mathbb{R}^{|\mathcal{E}_b| \times p(M)}$$

to have rank $p(M)$: at least $p(M)$ estimation vintages with linearly independent pricing-error sensitivities.

This puts a hard feasibility constraint on the block geometry: $|\mathcal{E}_b| = V - d - |\mathcal{G}_b|$ must comfortably exceed $p(M)$, and—because adjacent vintages have highly correlated factor paths—the *effective* rank of J_b can be lower than its row count. The block length d therefore trades off discrimination (R2 wants d large) against estimability (identification wants $\mathcal{E}_b$ rich); the pre-registered baseline $d \approx 0.2 |\mathcal{V}^S|$, $g = 1$ leaves roughly three-quarters of the vintages in estimation. The relevant empirical check is not a moment count but the rank or condition number of the estimated Jacobian at $\hat{\theta}_b$, reported per block alongside cross-block parameter dispersion and boundary frequencies. Note also that the unseasoned Euler vintages, though never scored, contribute estimation-side rank: they extend the factor-path coverage to the most recent calendar years.

4.5 Block averaging as subagging

Averaging over the enumerated blocks is subsample aggregation—*subagging*—of the block-level estimator [Bühlmann and Yu, 2002], with the vintage subsets as the subsamples. The framework reports two bagged objects: the bagged parameter $\bar{\theta} = \frac{1}{B} \sum_b \hat{\theta}_b$ and the score Score. Bagging smooths the dependence of the nonlinear argmin on the vintage composition and reduces its variance, with the largest gains for unstable estimators—exactly the regime of small PE panels with occasional boundary or near-degenerate solutions. The cross-block dispersion of $\hat{\theta}_b$ is itself informative: it is a delete- d -jackknife-style stability measure taken at the vintage level, which is the correct cluster level for PE dependence. What block averaging cannot do is create identification: if the block-level moments were rank deficient, averaging many rank-deficient solutions would produce a tight bag around a pseudo-value—false precision (Appendix A). Subagging is valuable on top of the pooled vintage-vector identification, not as a substitute for it.

4.6 Inference for score differences

Model rankings need uncertainty measures. For two model-design pairs a and b , compute their scores on the same block and form the paired difference $d_b(a, b) = S_b(a) - S_b(b)$. Pairing matters because both models face the same estimation vintages, the same held-out vintages, and the same factor realizations, which removes a large amount of design noise from the comparison.

The naive standard error S_d^2/B treats the B paired differences as independent. They are not: the blocks overlap heavily in their estimation sets, and neighboring blocks share held-out vintages, so $\text{Cov}(d_b, d_{b'}) > 0$. The genuine uncertainty lives in the finite, dependent cross-section of vintages and funds and does not shrink with the number of evaluation windows—the cross-validation variance problem of Bengio and Grandvalet [2004], who show that no unbiased estimator of the variance of a resampled score exists. Three complementary responses:

- *Corrected t* . The correction of Nadeau and Bengio [2003] inflates the naive variance by the held-out-to-estimation ratio,

$$\widehat{\text{Var}}_{\text{corr}}(\bar{d}) = \left(\frac{1}{B} + \frac{d}{|\mathcal{E}_b \cap \mathcal{V}^S|} \right) S_d^2$$

which converges to a positive limit rather than to zero as evaluation windows accumulate. Because the enumerated blocks are ordered in time, a heteroskedasticity-and-autocorrelation-robust variance over the block sequence is a natural companion; agreement between the two is reassuring.

- *Cluster bootstrap.* The more faithful alternative resamples at the dependence-relevant cluster level—vintages, or GP families within vintages—and reruns the entire pipeline (blocks, estimation, scores) per bootstrap draw, taking the distribution of $\bar{d}^*(a, b)$ as the sampling distribution of the score gap.
- *Multiplicity.* Because the goal is to rank several models, per-pair tests are embedded in a model-confidence-set procedure [Hansen et al., 2011]: we report the set of statistically indistinguishable top models rather than a single winner whose lead is within the corrected standard error.

Boundary-alpha solutions are reported as a paired diagnostic as well: if one model wins only because it frequently lands on the alpha boundary, the ranking is not economically interpretable.

5 Unseasoned Vintages: A Hybrid Cash-Flow/Euler Estimator

5.1 The dilemma

Unseasoned vintages carry valuable pricing information, but their value is still locked in appraised NAVs. The standard treatment—discounting the latest NAV back to fund inception as a terminal cash flow—creates two problems. First, it concentrates a large share of the fund’s discounted value at a single calendar date, the dataset endpoint, so the estimator’s fit becomes disproportionately sensitive to the macro state at that date rather than to the fund’s lifetime factor exposure. Second, the terminal NAV is an appraisal, not a realized payoff; its co-movement with public indices may differ systematically from that of actual divestitures. Excluding recent vintages wastes data; including them naively buys fragile, NAV-driven estimates.

Our resolution keeps realized cash flows wherever they exist and confines NAV information to short horizons: seasoned vintages contribute the cumulative NPV moment g_i^S , unseasoned vintages contribute normalized one-period Euler errors on quarterly NAV returns. In the blocked scheme the division of labor is strict: unseasoned vintages support *estimation*—they extend factor-path coverage to the most recent years and add identifying rank—but they are never scored. Two alternative treatments—adjusting terminal NAVs with secondary-market discounts, and pricing NAV returns for the entire sample—are natural competitors; we develop them, and the design for racing all three, in Appendix C, and summarize the comparison in Section 5.4.

5.2 The normalized one-period Euler block

One-period Euler pricing error. For a fund i in an unseasoned vintage with quarterly NAV observations at dates $t = 0, 1, \dots, T_i$, define $R_{t,i}^H$ as the gross quarterly Horizon-IRR return that solves

$$\text{NAV}_{t-1,i} R + \sum_{\ell \in \mathcal{X}_{t,i}} X_{\ell,i} R^{1-\lambda_{\ell,t,i}} = \text{NAV}_{t,i} + \sum_{\ell \in \mathcal{D}_{t,i}} D_{\ell,i} R^{1-\lambda_{\ell,t,i}},$$

where $\mathcal{X}_{t,i}$ and $\mathcal{D}_{t,i}$ are the dated capital calls and distributions inside the quarter and $\lambda_{\ell,t,i} \in [0, 1]$ is the fraction of the quarter elapsed when cash flow ℓ occurs. If only quarter-end NAVs and aggregate quarterly net cash flows are available, the simple return $(CF_{t,i} + \text{NAV}_{t,i})/\text{NAV}_{t-1,i}$ is a coarse approximation, but the Horizon-IRR construction is preferred because it accounts for intra-quarter timing. The relative one-period

pricing error for quarter t is

$$e_{t,i}^H(\theta) = M_{t-1,t}(\theta) R_{t,i}^H - 1, \quad E[e_{t,i}^H(\theta_0) | \mathcal{F}_{t-1}] = 0,$$

with the same one-period SDF $M_{t-1,t}(\theta)$ as the seasoned block. The conditional statement matters: pre-terminated normalizations and instruments preserve the moment only under a conditional Euler restriction. Crucially, these quarterly errors are *not* discounted back to the initial date with the candidate cumulative SDF—doing so would algebraically reconstruct the discounted terminal-NAV term the block is designed to avoid (Appendix E).

Normalization and scale matching. Average the relative quarterly errors within fund,

$$\bar{r}_i(\theta) = \frac{1}{T_i} \sum_{t=1}^{T_i} e_{t,i}^H(\theta),$$

and rescale by a deterministic duration factor that maps the fund-average relative error into NPV-like units per committed dollar. Let $\bar{u}_s(a)$ be a benchmark age- a profile of NAV/ K for strategy s , and q_{a-1} a deterministic discount sequence (baseline $q_{a-1} = 1$). Define

$$\Lambda_i = c_{s(i)} \sum_{a=1}^{T_i} q_{a-1} \bar{u}_{s(i)}(a), \quad g_i^U(\theta) = \Lambda_i \bar{r}_i(\theta),$$

with $c_{s(i)} = 1$ in the baseline, which keeps Λ_i entirely outside the SDF and outside the pricing errors. The interpretation is simple: $\bar{r}_i$ is the average relative quarterly mispricing, and Λ_i is an externally chosen capital-at-risk duration—how much NAV/ K exposure a typical fund of that age and strategy carries over its observed life. If the relative quarterly mispricing is roughly stable, $e_{t,i}^H(\theta) \approx \rho_i(\theta)$, then $g_i^U(\theta) \approx \rho_i(\theta)\Lambda_i$, the same order as a life-of-fund NPV error per committed dollar. The pooled unseasoned vintage moment $\epsilon_v(\theta)$ is the average of $g_i^U(\theta)$ over the vintage’s funds, exactly as in the seasoned block. A deterministic-duration variant that weights the quarterly errors directly, $g_i^{U,\text{det}}(\theta) = \sum_t q_{t-1} \bar{u}_{s(i)}(t) e_{t,i}^H(\theta)$, serves as a robustness check on the scale matching.

Cross-fitting nuisance choices. Any nuisance component used to build the unseasoned moments—the threshold δ if tuned, the duration profile $\bar{u}_s(a)$, an optional scale constant c_s —must be selected without reference to held-out scores, and in rolling-origin designs using only information observable at the origin date. Pre-registering Λ_i from external NAV-profile curves, with no reference to pricing errors, eliminates the circularity concern entirely and is the baseline.

NAV quality and the Euler restriction. For traded assets, no-arbitrage forces the one-period Euler condition to hold; for PE funds, NAVs are appraisals whose co-movement with public factors may be smoothed or biased, so the restriction holds only approximately—and appraisal smoothing biases measured loadings in a predictable direction (toward zero). The hybrid design controls this concern in four ways. First, only unseasoned vintages rely on NAV-based moments at all; seasoned vintages are priced from realized cash flows. Second, even within $\mathcal{V}^U$ the block uses one-quarter NAV return changes, not a terminal NAV discounted over a decade. Third, no NAV-based moment is ever scored: the Euler block can tilt $\hat{\theta}$, but it cannot make a bad model look good on the reported criterion. Fourth, as the panel grows, vintages migrate from $\mathcal{V}^U$ to $\mathcal{V}^S$, so the estimator is increasingly cash-flow dominated. The simulation stresses the residual exposure directly by switching appraisal smoothing on and off (Section 7).

5.3 NAV-free evaluation

Because the hybrid estimator leans on appraisals for part of the sample, its evaluation must not. We state this as a design rule: *the metric used to compare models and designs never touches a NAV*. Realized contributions

and distributions are observed without appraisal error; they are the natural common currency for scoring. The blocked scheme satisfies the rule by construction—held-out blocks are drawn from seasoned vintages only—so the headline score is already NAV-free. Two further designs extend the rule to the questions the blocked scheme cannot answer.

The realized-fund target. Fix an evaluation date τ' and let $\mathcal{R}_{\tau'}$ be the funds that are realized as of τ' —residual NAV below the de minimis threshold. For such funds the entire life is observed in realized cash flows, so the NAV-free pricing error is the seasoned contribution evaluated on a now-complete fund,

$$g_i^{\text{real}}(\theta) = \frac{1}{K_i} \sum_{t=0}^{T_i} \Psi_{0,t}(\theta) CF_{t,i}, \quad i \in \mathcal{R}_{\tau'},$$

with any de minimis residual handled by the single terminal-NAV convention shared across all specifications. Vintage-level aggregation and U-statistic scoring proceed exactly as in Section 4.3, so specifications differ only in how their parameters were *estimated*, never in how they are *scored*.

Two designs that make the target informative. The target discriminates between treatments of unseasoned vintages only if the held-out realized funds were treated differently at estimation time.

- *Rolling origin (prospective).* Pick a historical information date τ . Using only information available at τ , classify then-seasoned and then-unseasoned vintages and estimate $\hat{\theta}_{\tau}$. Move the evaluation date forward until the then-unseasoned vintages have seasoned, and score $\hat{\theta}_{\tau}$ on their realized cash flows. This is the terminal-block member of the block family, with the information set also truncated: fully out-of-time, at the cost of being one-sided and feasible only for origins old enough that their young vintages have since realized.
- *Leave-future-out (retrospective).* Without rolling the information set back, truncate each currently-seasoned fund at an elapsed age $\bar{a}$, estimate on the truncated panel—where the treatments genuinely differ about the unrealized tail—and score on the realized cash flows after $\bar{a}$. This uses the full panel and is cheaper than rolling-origin re-estimation, at the cost of not being a real-time experiment.

A selection caveat. The realized-fund target conditions on funds that have realized by the evaluation date, so slow-to-liquidate funds are under-represented. Two safeguards keep the target honest: report the share of estimation-time funds that qualify (coverage), and rely on the *paired* comparison, in which every specification is scored on the same realized funds at the same origin, so the selection cancels in the cross-specification difference even if it biases the level.

Group-wise diagnostics. Although unseasoned vintages are never scored, the implementation reports their estimation-side fit separately. If the Euler moments are trivially satisfied at $\hat{\theta}_b$, they contribute little identifying content; if they are in sharp tension with the seasoned moments, the hybrid compromise—and possible appraisal bias—is doing real work, and the appraisal backtest of Appendix C.4 localizes where.

5.4 Alternative treatments

Two competitors to the hybrid block are developed in Appendix C. *Approach A* keeps the cumulative cash-flow SDF for all vintages but multiplies each unseasoned fund’s terminal NAV by an empirical secondary-market discount before discounting it to inception; it is conceptually simple and marks appraisals to transaction levels, but it retains the long-horizon terminal-NAV exposure and adds a discount-model nuisance. *Approach B* abandons cash-flow pricing entirely and prices quarterly NAV returns for *all* vintages, in the spirit of the NAV-return literature [Brown et al., 2023, Ghysels et al., 2025]; it is internally consistent and avoids long-horizon discounting, but it relies on appraisal quality even where realized cash flows exist. The hybrid

(Approach C) is the paper’s baseline because it uses realized cash flows wherever available and confines NAV reliance to short horizons on young funds. All three produce vintage-level pricing errors that plug into the same blocked machinery, and the NAV-free rolling-origin score of Section 5.3 is the common criterion that can rank them; Appendix C specifies the horse race.

6 Alpha Design

The resampling method does not by itself decide what alpha should mean. The baseline is the cumulative, NPV-style alpha already embedded in the SDF of Section 2: the cumulative discount factor $\Psi_{0,t}(\theta) = e^{-a\tau_t} / B_{0,t}(\beta)$ discounts by the factor benchmark and credits a continuously compounded abnormal rate a . This is the direct-alpha/GPME tradition [Gredil et al., 2023, Korteweg and Nagel, 2016, 2025]: alpha is a long-horizon abnormal-performance object, which is the natural quantity for PE cash flows; the discretely compounded $(1 + \alpha)^{-t}$ is equivalent up to $a = \log(1 + \alpha)$. The main alternative, retained for comparison in the design vector and discussed in Appendix F, is the per-period intercept alpha of Driessen et al. [2012], in which α sits inside the one-period factor denominator.

Vintage-implied alpha as a diagnostic. For fixed loadings $\hat{\beta}$, each vintage has a well-defined implied cumulative alpha: $a_v(\hat{\beta})$ solves the IRR-type equation

$$\sum_t \frac{cf_{v,t}}{B_{0,t}(\hat{\beta})} e^{-a\tau_{v,t}} = 0$$

on the vintage’s pooled cash-flow series, which has a single inception origin and, under a single sign change (calls before distributions), a unique root. In estimation, a common a is part of θ ; in diagnostics, the cross-vintage dispersion of $\{a_v(\hat{\beta})\}$ —and in particular the gap between held-out and estimation vintages’ implied alphas per block—measures how sensitive the alpha estimate is to which vintages are used. A large spread flags a weakly identified aggregate alpha.

No exploding-alpha degeneracy. A known finite-sample pathology of NPV-style SDF estimation, first documented by Driessen et al. [2012], is worth stating precisely because it is often described imprecisely. The problem is *not* the length of the discounting horizon per se. It is a limit: as $\alpha \rightarrow \infty$, every discounted cash flow is driven to zero, so each fund’s pricing error converges to its *undiscounted* cash flows—the flows at elapsed time zero. If the moment is constructed so that these undiscounted flows are small (for example, sums indexed from $t = 1$, or tiny first calls), the squared loss along the large- α direction approaches zero and a ridge of spurious near-minima appears. Two design features block this, and it is the *indexing* that does the heavy lifting: with time-zero indexing, the undiscounted anchor is the full initial capital call per committed dollar, so the loss along the ridge is bounded below by the squared anchor—far above the interior minimum—under either alpha design. The cumulative parameterization adds separability (alpha multiplies benchmark-discounted flows, so the vintage-implied alpha is a unique interior IRR-type root rather than a boundary escape), and the unseasoned Euler block reinforces the discipline from the other side: as $\alpha \rightarrow \infty$ the one-period SDF collapses and $e_{t,i}^H \rightarrow -1$, the maximal relative pricing error, so extreme alpha is actively penalized rather than rewarded. We therefore need only two cheap safeguards in the empirical work—a pre-registered bounded alpha box and a reported frequency of boundary solutions—and refer the full anatomy of the degeneracy, together with identification-robust (Anderson–Rubin-type) confidence sets for alpha, to Appendix B.

7 Semi-Synthetic Simulation Evidence

Three properties of the framework cannot be established on real data alone: whether the criterion selects the *true* factor model (selection power and size), how biased the estimated alpha and loadings are in panels of this size, and whether the corrected tests have honest size. A compact semi-synthetic simulation delivers all three while staying as close to the actual empirical problem as possible: everything observable about the design is held *real*, and only the unobservable—the data-generating SDF—is synthetic.

Design. The simulation fixes, from the buyout panel of Section 8: the historical factor and risk-free paths; the vintage years; the number of funds and commitment sizes per vintage; and empirical age profiles of capital calls and exit timing. Synthetic cash flows are then generated under a known SDF: fund i of vintage v calls capital along the empirical call profile; each called dollar exits after a holding period drawn from the empirical exit-age distribution; and exit proceeds compound the benchmark at the true parameters plus idiosyncratic noise,

$$D = X \cdot \exp\left(\sum_{s \in \text{holding period}} \left[a_0 \Delta_s + \log(1 + r_{f,s} + \beta'_0 f_s) \right] + \sigma \varepsilon_i - \frac{1}{2} \sigma^2 \right), \quad \varepsilon_i \sim N(0, 1),$$

so the true risk-adjusted NPV is zero at (a_0, β_0) by construction. Unseasoned funds' NAVs are model-consistent continuation values, optionally passed through an AR(1) smoothing filter with parameter ϕ to mimic appraisal smoothing. True specifications vary over an alpha-only model, a one-factor model, and a two-factor model, with empirically plausible (a_0, β_0, σ) ; the candidate set is the design-table model list.

Reported experiments. Four experiments, one table each:

1. *Selection.* Frequency with which each candidate is selected—and the model confidence set covers the truth—under (i) blocked vintage cross-validation, (ii) within-vintage fund-level cross-validation, and (iii) in-sample GMM objective comparison. The central prediction of Section 4.2: as fund counts grow, (ii) drifts toward the largest candidate while (i) concentrates on the truth. **[TBD: table]**
2. *Estimation.* Bias and RMSE of $\hat{a}$ and $\hat{\beta}$ at the panel's actual size; boundary frequencies; distribution of Jacobian condition numbers across blocks. **[TBD: table]**
3. *Inference.* Empirical size of nominal-5% paired tests using the naive resampled t , the corrected t , and the vintage-cluster bootstrap. **[TBD: table]**
4. *Appraisal stress.* Sensitivity of $\hat{\theta}$ and of the seasoned-only score to the smoothing parameter ϕ and to the seasoning threshold δ , quantifying how much protection the never-score-a-NAV rule actually buys. **[TBD: table]**

Block-geometry sensitivity (d, g) is reported alongside experiment 1. The simulation deliberately conditions on the single realized factor history, exactly as the estimator does; it therefore measures uncertainty from fund-level idiosyncrasies and appraisal behavior, not factor-path uncertainty, and its scope should be read accordingly.

8 Empirical Application: Buyout Funds

The empirical implementation applies the framework to a single-strategy buyout sample. Restricting to one strategy keeps the vintage moment vector homogeneous, avoids strategy-mix confounds, and is where fund counts per vintage are largest.

8.1 Data

[TBD: Buyout fund sample: source, coverage, filters. Required fields: fund-level quarterly contributions, distributions, and NAVs, net of fees; committed capital; vintage year. Target panel: seasoned vintages roughly 1999–2014, unseasoned vintages thereafter, classified by the unrealized-value fraction with $\delta = 0.15$ at the estimation date. Report funds per vintage, aggregate commitments, and the seasoned/unseasoned boundary implied by δ .]

8.2 Design

Table 1 pre-registers the design space. All candidate models are estimated and scored on the identical enumerated block sequence, so all comparisons are paired.

Component	Specification (baseline first)
SDF models M	risk-free only; market (one factor); levered market; [TBD: final list]
Alpha design	cumulative alpha (baseline); DLP12 intercept alpha (comparison only)
Moment family	hybrid NPV/Euler (baseline); pooled PME on $\mathcal{V}^S$ (robustness)
Unseasoned treatment	Euler block in estimation, never scored (baseline); exclude $\mathcal{V}^U$ (robustness)
Seasoning threshold	$\delta = 0.15$; $\delta \in \{0.10, 0.20\}$ (robustness)
Vintage weights	equal (baseline); value weights (robustness)
Holdout scheme	blocked vintage CV: $d = \lceil 0.2 \mathcal{V}^S \rceil$, $g = 1$, enumerated (baseline); d, g variations and random vintage subsets (robustness); within-vintage fund CV (stability diagnostic only)
Scoring	U-statistic $\widehat{U}_v$, seasoned held-out vintages only; raw ϵ_v^2 as companion
Admissible set Θ_M	benchmark-denominator positivity on observed support; $a \in [\underline{a}, \bar{a}]$ pre-registered

Table 1: Pre-registered design space for the buyout application.

8.3 Algorithm

For each candidate model M and design η :

1. Classify vintages into $\mathcal{V}^S$ and $\mathcal{V}^U$ at the estimation date; fix nuisance objects (duration profiles $\bar{u}_s$, terminal-NAV convention) from pre-registered external inputs.
2. Enumerate the admissible blocks: contiguous windows of length d in $\mathcal{V}^S$, each with buffer g ; verify rank feasibility of every estimation set (condition number of the estimated Jacobian).
3. For each block b : build pooled vintage moments on $\mathcal{E}_b$; estimate $\hat{\theta}_b$ over Θ_M ; score the held-out vintages with $\widehat{U}_v(\hat{\theta}_b)$; record convergence, boundary, and conditioning diagnostics.
4. Aggregate: Score, bagged $\bar{\theta}$, cross-block parameter dispersion, per-vintage holdout coverage.
5. Rank model-design pairs by Score; report the model confidence set under corrected and bootstrap inference.

Alongside the mean held-out loss, we report: the median and dispersion of block scores; raw ϵ_v^2 companions to $\widehat{U}_v$; the scalar-aggregate cancellation diagnostic of Section 3.3; estimation-side seasoned/unseasoned fit; vintage-implied-alpha dispersion; and the fund-level stability diagnostic (dispersion of $\hat{\theta}$ under within-vintage fund resampling, Figure 2).

8.4 Main results

[TBD: Main ranking table: candidate models $\times$ {Score, corrected SE of paired difference to best model, model-confidence-set membership, boundary frequency, median Jacobian condition number}, under the baseline design. Headline questions: (i) which factor structure prices held-out buyout vintages best; (ii) how large is the estimated cumulative alpha in the preferred model, with its identification-robust confidence set from Appendix B as a companion; (iii) how stable are the loadings across blocks.]

8.5 Out-of-time validation of the unseasoned block

[TBD: Rolling-origin exercise: estimate at origins τ (e.g., annual origins over 2012–2020) under (i) the hybrid baseline and (ii) the exclusion benchmark; score on the NAV-free realized-fund target of Section 5.3 once the then-unseasoned vintages have seasoned; report paired differences with corrected standard errors and target coverage. If the data supports it, add Approaches A and B from Appendix C to the race.]

8.6 Implementation and placebo checks

Alpha-injection invariance check. Tilting every observed cash flow and NAV by a known abnormal rate, $\widetilde{CF}_{t,i} = CF_{t,i} e^{a_0 \tau_t}$, shifts the pricing-consistent alpha by exactly a_0 and leaves the loadings and all pricing errors unchanged: the tilted panel satisfies every moment at $(a + a_0, \beta)$ if and only if the original satisfies it at (a, β) . Because the invariance is exact, this is an *implementation* check, not a statistical one: re-running the pipeline on tilted panels verifies the optimizer, the moment code, and the placement of the pre-registered alpha box (which is the only element that can break the invariance), while statistical recovery evidence comes from the simulation of Section 7. [TBD: one-line result.]

Factor-scramble placebo. Destroy the alignment between cash flows and factor realizations while preserving both marginals: replace the factor path with a calendar-shifted (or block-permuted) path and re-run the pipeline. Factor loadings can then no longer reflect genuine risk adjustment, so the held-out advantage of factor models over the alpha-only benchmark should collapse toward zero across blocks. A model that still “wins” on scrambled factors is winning through design overfitting, not pricing ability; a ranking that collapses under the placebo and is strong on the real alignment is evidence that the blocked score responds to genuine factor–cash-flow covariation. [TBD: placebo table.]

9 Conclusion

Private-equity SDF estimation has always involved consequential moment-construction choices; this paper makes them explicit and, where possible, derives them. The resampling design itself is derived from three finite-sample requirements—pool the exchangeable, hold out the identifying variation, score without bias—yielding blocked vintage cross-validation with pooled moments and unbiased U-statistic scoring, enumerated exhaustively so that no Monte Carlo tuning remains. Around this core, the paper commits to design choices that remove two chronic pathologies at the root—a hybrid cash-flow/Euler treatment under which no appraisal is ever discounted over a long horizon or admitted into a comparison score, and a cumulative alpha that is anchored against the exploding-alpha degeneracy—and ranks what remains, the SDF models themselves, by their ability to price buyout vintages they have not seen. A semi-synthetic simulation verifies that the criterion can find true factor structure where fund-level cross-validation cannot, and the buyout application puts the

framework to work. Extensions the framework accommodates but this paper does not pursue—vintage-stratified and age-varying factor exposures, alternative alpha definitions, and richer moment weighting—are collected as a research agenda in Appendix F.

A Identification and Pooling: Formal Analysis

This appendix formalizes the two structural claims behind requirement R1 of Section 4.2: stacked fund-level folds are redundant moments (rank), and randomized sub-group moments are dominated by the pooled moment (Rao–Blackwell).

A.1 Exchangeable folds are rank deficient

Consider the (rejected) fold estimator that partitions each vintage’s funds into cells $C_{v,1}, \dots, C_{v,P}$, collapses each fold r to the aggregate $m_r(\theta) = \sum_v \omega_v \epsilon_{v,r}(\theta)$ with $\epsilon_{v,r}$ the cell mean, and stacks the P folds as a GMM system.

Within a vintage, the cells are a random partition of the same funds, so each cell mean is an unbiased draw of the vintage population pricing error:

$$\bar{g}_v(\theta) = E[g_i(\theta) \mid i \in \text{vintage } v], \quad E[\epsilon_{v,r}(\theta)] = \bar{g}_v(\theta) \quad \text{for every } r.$$

Hence every fold has the same population moment, $E[m_r(\theta)] = \sum_v \omega_v \bar{g}_v(\theta) =: \bar{m}(\theta)$ for $r = 1, \dots, P$. Stacking the P fold moments into $G(\theta) = \bar{m}(\theta) \mathbf{1}_P$ yields a Jacobian that is an outer product,

$$\frac{\partial G(\theta_0)}{\partial \theta'} = \mathbf{1}_P \frac{\partial \bar{m}(\theta_0)}{\partial \theta'}, \quad \text{rank}\left(\frac{\partial G(\theta_0)}{\partial \theta'}\right) \leq 1.$$

The local identification (rank) condition fails for any $p(M) \geq 2$, no matter how many folds are used: the P folds restrict only the single scalar combination $\bar{m}(\theta) = 0$ and leave a $(p(M) - 1)$ -dimensional set of observationally equivalent parameters. A pricing moment is *nonredundant* only if its population pricing error is not a linear combination of the others and its derivative with respect to θ adds independent rank; replicated exchangeable folds are duplicates on both counts.

The redundancy is structural, not a feature of any particular SDF. Folds are exchangeable, so they share one population moment and one Jacobian row. Vintages are not exchangeable: each vintage lives through a different factor path and cash-flow timing, so the sensitivities $\partial \bar{g}_v(\theta_0)/\partial \theta'$ differ across v . The identifying rank lives in the cross-vintage Jacobian

$$J(\theta_0) = \frac{\partial}{\partial \theta'} (\bar{g}_v(\theta_0))_v \in \mathbb{R}^{V \times p(M)}, \quad \text{identification} \iff \text{rank } J(\theta_0) = p(M).$$

Collapsing across vintages replaces J by its column sum $\mathbf{1}'J$, a single row, discarding exactly the cross-vintage rank that identifies the factor loadings. This is why estimation operates on the pooled vintage moment *vector*.

Weak identification and diagnostics. In finite samples the scalar fold moments differ only through sampling noise in cell composition, so the between-fold Jacobian variation is $O(1/|C_{v,r}|)$ and vanishes as cells grow: the fold estimator is weakly identified in the sense of Stock and Wright [2000], and its apparent finite-sample precision is an artifact of that noise. Because even many stacked moments can be rank deficient, the relevant empirical check is the rank or condition number of the estimated moment Jacobian at $\hat{\theta}$, not the moment count; it is reported per block in Section 8.3.

A.2 Pooling as Rao–Blackwellization

Fix θ and vintage v with n_v funds, and let $\epsilon_v(\theta)$ be the pooled mean of $g_i(\theta)$. Under a uniform random partition into cells of size $|C|$, each cell mean is a simple random sample without replacement, so

$$\mathbb{E}_\Pi[\epsilon_{v,r}(\theta)] = \epsilon_v(\theta), \quad \text{Var}_\Pi[\epsilon_{v,r}(\theta)] = \frac{\widehat{\sigma}_v^2(\theta)}{|C|} \left(1 - \frac{|C| - 1}{n_v - 1}\right),$$

where $\widehat{\sigma}_v^2(\theta)$ is the within-vintage cross-sectional variance and the expectation is over partitions holding the data fixed. Cell-based moments are therefore mean-preserving spreads of the pooled moment. For any criterion Q that is convex in the moment vector at fixed θ (all quadratic losses used here), Jensen’s inequality gives

$$\mathbb{E}_\Pi[Q(\epsilon_\Pi(\theta))] \geq Q(\epsilon(\theta)),$$

with equality only for trivial partitions: randomizing inside the moments inflates every convex objective in expectation, and the pooled moment vector is the Rao–Blackwellization of any cell-based construction. The same decomposition exhibits the fund-noise floor removed by the U-statistic score of Section 4.2: $E[\epsilon_v(\theta)^2 \mid \text{path}] = \bar{g}_v(\theta)^2 + \sigma_v^2(\theta)/n_v$, while $E[\widehat{U}_v(\theta) \mid \text{path}] = \bar{g}_v(\theta)^2$ exactly.

Subagging cannot create rank. Averaging estimators over subsamples stabilizes them; it does not create identification. If the base estimator is built from exchangeable fold moments, its Jacobian is rank deficient and averaging many such solutions only produces a tight bag around a partition-induced pseudo-value—false precision. Block averaging (Section 4.5) is valuable on top of the pooled vintage-vector identification, not as a substitute for it.

B The Exploding-Alpha Degeneracy

This appendix records the anatomy of the exploding-alpha pathology and inference tools for specifications that are exposed to it (notably the DLP12 intercept design retained as a comparison in Table 1). Under the baseline design the pathology is disciplined by construction (Section 6); the material here is the referee-proof backstop.

B.1 Anatomy

Mechanism. The degeneracy is a limit, not a horizon effect. As $\alpha \rightarrow \infty$, every discounted cash flow is driven to zero, so each fund’s pricing error converges to its undiscounted cash flows—the flows at elapsed time zero. The squared loss along the large- α direction therefore approaches the squared undiscounted anchor. Spurious near-minima arise exactly when that anchor is small: moment sums indexed from $t = 1$, gradual capital calls with tiny first flows, or origins placed before meaningful flows. In the intercept parameterization $M_{t-1,t} = 1/(1 + \alpha + \beta' f_t)$ the collapse direction additionally kills all β -sensitivity, producing a ridge along which the objective is nearly flat in every coordinate [Driessen et al., 2012, 2008].

The initial-contribution anchor. When the time-zero capital call enters the cumulative moment undiscounted ($\Psi_{0,0} = 1$), driving $\alpha \rightarrow \infty$ sends the seasoned NPV moment to $-X_{0,i}/K_i \neq 0$ rather than to zero, so the squared loss is bounded below by the squared anchor and the ridge is self-limiting. This is why the paper indexes seasoned cash flows from $t = 0$ throughout, and why the magnitude of the early-call anchor is a sample-specific diagnostic for exposure to the pathology.

Population uniqueness. The pathology is a finite-sample phenomenon. Under increasing-domain asymptotics indexed by vintage years, the cross-vintage moment averages over many vintage realizations of the cumulative SDF. These realizations are not independent—adjacent vintages overlap in calendar time and share factor paths—so the relevant condition is weak dependence (mixing) across vintages, and the limit should be read as a thin asymptotic (one new vintage per year, perhaps thirty to forty in total). Subject to this caveat, the population criterion is uniquely minimized at θ_0 under standard conditions: no single alpha value can simultaneously zero out the discounted cash flows of many vintage realizations with heterogeneous timing and factor paths.

Safeguards in the framework. Four mechanisms limit the pathology in finite samples: (i) the time-zero anchor above; (ii) the unseasoned Euler block, whose one-period errors satisfy $e_{t,i}^H \rightarrow -1$ as $\alpha \rightarrow \infty$ —the maximal relative error—so extreme alpha is penalized, not rewarded; (iii) economically admissible restrictions imposed before scoring (benchmark-denominator positivity over the observed factor support, a pre-registered bounded alpha box); and (iv) cross-block diagnostics—weakly identified specifications display high parameter dispersion across blocks, frequent boundary solutions, and non-convergence. Held-out scoring alone is *not* a cure: a degenerate cumulative SDF that collapses all long-horizon discounted cash flows can make held-out NPV moments small as well.

B.2 Identification-robust inference for α

Where the ridge is potentially active, Wald-type confidence intervals for α are unreliable precisely when they are needed. The natural tools are weak-identification-robust statistics.

A profiled robust confidence set. Treat β as nuisance and profile it out at each hypothesized α : for a grid of values α_0 , refit the remaining parameters on the estimation vintages and evaluate the continuously-updated GMM objective $T(\alpha_0) = \min_{\beta} Q(\alpha_0, \beta)$. Under $H_0 : \alpha = \alpha_0$ an Anderson–Rubin / Stock–Wright S -statistic is asymptotically χ^2 regardless of identification strength [Stock and Wright, 2000], so inverting it,

$$CS_{1-a}(\alpha) = \{\alpha_0 : T(\alpha_0) \leq c_{1-a}\},$$

yields a confidence set with correct coverage even on the ridge. The payoff is diagnostic as much as inferential: when α is weakly identified the set is wide or unbounded above, displaying the ridge honestly, whereas a Wald interval would report spurious precision. The profiled curve $T(\cdot)$ pairs with the vintage-implied-alpha dispersion of Section 6: a flat right tail of T and a large spread of $\{\alpha_v\}$ are two readings of the same weak identification.

A PME anchor that breaks the degeneracy. The NPV moment loses its grip on α in the large- α region because its sensitivity $\partial m^{\text{NPV}} / \partial \alpha$ collapses as the cumulative discount factors vanish. The pooled-PME moment of Appendix D does not share this defect: being a ratio, it is invariant to a common rescaling of all discount factors and responds instead to the *relative* reweighting across maturities that α controls, so its α -sensitivity does not vanish on the ridge. Stacking a single pooled-PME moment (restricted to $\mathcal{V}^S$, where it is well defined) alongside the NPV moments keeps the α -column of the moment Jacobian away from zero and locally re-identifies α , while the linear NPV moments continue to do the efficient work for β .

C Alternative Treatments of Unseasoned Vintages

This appendix develops the two competitors to the hybrid Euler block of Section 5.2 and the design that races all three on a common criterion.

C.1 Approach A: NAV-adjusted cash-flow SDF

Adjust the terminal NAV before applying the standard cumulative cash-flow SDF. For seasoned vintages, the pricing restriction is unchanged. For unseasoned vintages, the latest reported NAV is multiplied by an empirical discount factor before being treated as a terminal distribution:

$$\widetilde{D}_{T_i,i} = D_{T_i,i} + d_i \cdot \text{NAV}_{T_i,i}, \quad d_i \in (0, 1],$$

after which the fund enters the standard CF-SDF pricing restriction with $\widetilde{D}_{T_i,i}$ as terminal cash flow. The discount d_i can be calibrated from secondary-market transaction data—for example, the Setter Capital secondary transaction dataset, which records bid–ask spreads and transaction prices for LP interests as a function of fund age, strategy, and market conditions—grounding the adjustment in observed prices rather than assumptions.

Advantages. Conceptually simple; one cumulative SDF for all vintages; the discount directly addresses appraisal bias by marking the terminal position to secondary-market levels.

Limitations. The discounted NAV is still a terminal cash flow discounted back to inception, so the estimator retains its sensitivity to the macro state at the dataset endpoint. The discount function d_i introduces model risk: secondary-market data may not represent the broader PE universe, and d_i may vary with market conditions in ways that interact with the SDF specification. As a nuisance object, d_i must be pre-registered from external curves or cross-fitted (never tuned against scores; chronological in rolling-origin designs).

C.2 Approach B: NAV-return SDF for all vintages

Abandon the cash-flow pricing restriction entirely and price quarterly Horizon-IRR NAV returns for *all* vintages, seasoned and unseasoned alike, using the one-period Euler error $e_{t,i}^H(\theta)$ of Section 5.2; the fund-level moment is the time-series average of quarterly errors, aggregated to pooled vintage moments as usual. This is closest in spirit to the NAV-return literature: Brown et al. [2023] nowcast fund NAVs, and Ghysels et al. [2025] construct high-frequency PE returns from NAV-based state-space models.

Advantages. Internally consistent: one restriction for all vintages, no seasoned/unseasoned partition, no NAV discount model. Each quarterly error depends only on the one-period SDF, so long-horizon discounting instabilities never arise.

Limitations. NAV quality matters for the *entire* sample, including seasoned vintages where realized cash flows are available and arguably preferable: divestiture cash flows co-move genuinely with public factors, appraisals may co-move in a smoothed or biased way. The approach trades the unseasoned-vintage dilemma for full-sample appraisal reliance.

C.3 Racing the approaches on a common criterion

Comparing held-out losses *across* approaches is not meaningful directly: the approaches define different moment conditions with different scales, economic content, and parameter sensitivities. The resolution is the NAV-free principle of Section 5.3: estimate $\hat{\theta}^{(k)}$ under each approach $k \in \{A, B, C\}$ using its own moments, but score all of them with the identical realized-cash-flow map. Within the blocked scheme, the common criterion is the U-statistic score of Section 4.3 applied to seasoned held-out vintages,

$$S_b^{\text{common}}(k) = \frac{1}{d} \sum_{v \in \mathcal{H}_b} \widehat{U}_v(\hat{\theta}_b^{(k)}), \quad k \in \{A, B, C\},$$

which is identical across approaches and free of NAVs. The sharper prospective version is the rolling-origin score: estimate at a historical date τ , when the approaches genuinely disagree about the then-unseasoned vintages $\mathcal{V}_\tau^U$, and score later on the realized cash flows of exactly those vintages once they have seasoned,

$$S_\tau^{\text{roll}}(k) = \frac{1}{|\mathcal{V}_\tau^U \cap \mathcal{V}_{\tau+\ell}^S|} \sum_{v \in \mathcal{V}_\tau^U \cap \mathcal{V}_{\tau+\ell}^S} \widehat{U}_v^{\text{real}}(\hat{\theta}_\tau^{(k)}),$$

where $\widehat{U}_v^{\text{real}}$ is the U-statistic built from the realized-fund errors g_i^{real} of Section 5.3. Every nuisance object (the NAV discount d_i , duration profiles, thresholds) must use only information observable at τ . The selection caveat and its safeguards (coverage reporting; paired differences on identical realized funds) are as in Section 5.3.

C.4 Backtesting the appraisal

The same realized continuations support a direct, dollar-denominated diagnostic of the appraisal-bias concern that motivates the entire problem. At a freeze date with elapsed quarters s , the model-implied value of the fund's later-realized cash flows is

$$\widehat{\text{NAV}}_{s,i}(\theta) = \frac{1}{\Psi_{0,s}(\theta)} \sum_{t>s} \Psi_{0,t}(\theta) CF_{t,i},$$

the cumulative-SDF price of the realized tail, renormalized to date s . The appraisal gap $\text{NAV}_{s,i} - \widehat{\text{NAV}}_{s,i}(\theta)$ then has a benchmark term built only from realized flows. This is a diagnostic rather than a comparison metric—it asks whether a fitted SDF rationalizes reported NAVs as fair risk-adjusted values, and it localizes *where* (which strategies, ages, or vintages) appraisals and model values diverge, which is exactly the smoothing and bias that Approaches A and B are most exposed to.

D Detailed Comparison of Pricing Moments

This appendix compares the dollar NPV and PME-ratio moment families. The main text uses the (hybrid) NPV moment as the baseline; the PME-ratio serves as a robustness check on $\mathcal{V}^S$.

Dollar NPV moment. For seasoned funds of vintage v , aggregate discounted cash flows into one cumulative pricing error,

$$m_v^{\text{NPV}}(\theta; M, \eta) = \sum_{i \in \mathcal{I}_v} w_i \sum_{t=0}^{T_i} \Psi_{0,t}(\theta; M, \eta) CF_{t,i}, \quad CF_{t,i} = D_{t,i} - X_{t,i},$$

where $X_{t,i}$ are contributions, $D_{t,i}$ are distributions, any small residual terminal NAV is handled according to the design convention in η , and $\Psi_{0,t}$ is the cumulative SDF. At the population parameter, $E[m_v^{\text{NPV}}(\theta_0)] = 0$: the risk-adjusted NPV of fund cash flows vanishes.

Pros. (i) *Linearity* in cash flows yields tractable GMM derivatives, influence functions, and CLT-based inference. (ii) *Additivity* across funds makes the pooled-moment and block structure of Section 8.3, as well as the U-statistic score and the bootstrap, transparent: the vintage moment is the mean of fund-level moments. (iii) *Compatibility with the normalized Euler NAV restriction* for unseasoned vintages: the seasoned block

remains a linear cash-flow NPV, while the unseasoned block uses lagged-scale one-period Euler errors rescaled by an exogenous duration factor. Both blocks are expressed as per-committed-dollar pricing errors, and neither requires ratio denominators that can approach zero.

Cons. (i) *Scale-dependence.* A \$1B fund with a 1% relative NPV error contributes ten times more to the moment than a \$100M fund with the same relative error, so efficient GMM weighting implicitly tilts the estimator toward mega-funds. The influence function is mechanically heavy-tailed in the fund-size distribution rather than in genuine pricing disagreement. (The per-committed-dollar normalization $w_i = 1/K_i$ of the main text removes this tilt at the fund level.) (ii) *Dollar-denominated comparisons across vintages* with heterogeneous aggregate commitments are not directly meaningful; normalization is left to the econometrician. (iii) *Sign cancellation.* A cross-sectional mean of zero is consistent with systematic under-pricing of one subgroup offsetting over-pricing of another, which the scalar moment cannot detect. This motivates the vintage-vector construction of Section 3.3 but does not eliminate the concern within a vintage.

PME-ratio moment. Define the risk-adjusted PME of fund i as

$$\text{PME}_i(\theta; M, \eta) = \frac{\sum_t \Psi_{0,t}(\theta; M, \eta) D_{t,i}}{\sum_t \Psi_{0,t}(\theta; M, \eta) X_{t,i}},$$

from which two natural vintage-level moments follow. The *average-PME* form is

$$m_v^{\text{PME}}(\theta) = n_v^{-1} \sum_{i \in \mathcal{I}_v} (\text{PME}_i(\theta) - 1),$$

and the *pooled-PME* form is

$$m_v^{\text{PME-pool}}(\theta) = \frac{\sum_{i,t} \Psi_{0,t}(\theta) D_{t,i}}{\sum_{i,t} \Psi_{0,t}(\theta) X_{t,i}} - 1.$$

These two objects have different statistical meanings. Let

$$D_i^*(\theta) = \sum_t \Psi_{0,t}(\theta) D_{t,i}, \quad X_i^*(\theta) = \sum_t \Psi_{0,t}(\theta) X_{t,i}, \quad N_i^*(\theta) = D_i^*(\theta) - X_i^*(\theta).$$

The dollar-NPV restriction is $E[N_i^*(\theta_0)] = 0$. The average-PME population moment is

$$E\left[\frac{D_i^*(\theta_0)}{X_i^*(\theta_0)} - 1\right] = E\left[\frac{N_i^*(\theta_0)}{X_i^*(\theta_0)}\right] = \text{Cov}\left(N_i^*(\theta_0), \frac{1}{X_i^*(\theta_0)}\right),$$

where the last equality uses $E[N_i^*(\theta_0)] = 0$. Thus NPV pricing does not generally imply that the average PME has mean one; it does so only under stronger conditions, for example $E[N_i^*(\theta_0) \mid X_i^*(\theta_0)] = 0$. By contrast, the population pooled-PME object is

$$\frac{E[D_i^*(\theta_0)]}{E[X_i^*(\theta_0)]} - 1 = \frac{E[N_i^*(\theta_0)]}{E[X_i^*(\theta_0)]},$$

so it is aligned with the dollar-NPV restriction at the population level provided $E[X_i^*(\theta_0)] > 0$. The sample pooled PME is still a random ratio and can be biased in small samples, but it converges to a ratio of means rather than to the mean of individual ratios.

Pros. (i) *Scale-invariance.* The ratio is unit-free, so fund-size heterogeneity does not mechanically up-weight mega-funds. A \$100M fund and a \$1B fund enter the cross-sectional moment on equal footing.

(ii) *Practitioner interpretability*. PME is the native performance metric of the industry; reporting results in the same units aids dissemination. (iii) *Bounded downside*. The fund-level PME is bounded below by zero, so $\text{PME}_i - 1$ is bounded below by -1 : a single disaster fund cannot dominate the moment the way it can dominate a dollar NPV.

Cons. (i) *Nonlinearity*. The moment is a ratio in θ ; its derivative involves the ratio of two correlated sums and is numerically fragile when the denominator (discounted contributions) is small. This occurs mechanically for young funds with back-loaded capital calls — precisely the vintages that the Euler NAV restriction is designed to retain. (ii) *Mean-of-ratios versus ratio-of-means*. The average-PME and pooled-PME moments need not agree even asymptotically. Average PME estimates $E[D_i^*/X_i^*]$, while pooled PME estimates $E[D_i^*]/E[X_i^*]$. The gap is an economic weighting choice, not a minor finite-sample nuisance: average PME gives each fund equal weight, whereas pooled PME weights funds by discounted contributions. (iii) *No clean short-horizon analogue for unseasoned vintages*. Integrating unseasoned vintages $\mathcal{V}^U$ into a PME framework typically requires treating the latest $\text{NAV}_{T,i}$ as a terminal distribution and discounting it back to the fund inception date — exactly the long-horizon discounting mechanism that the normalized Euler block is designed to avoid. The PME robustness check is therefore restricted to $\mathcal{V}^S$. (iv) *Inference*. Upper-tail skewness in PME_i (funds can return many times their capital, but cannot lose more than it) distorts CLT-based standard errors and t -statistics unless explicitly robustified, e.g., by bootstrapping the vintage-level moment or by employing median-based loss functions.

E Why We Do Not Discount Euler Errors to Time Zero

Let

$$\eta_{t,i}^{\$}(\theta) = M_{t-1,t}(\theta)(CF_{t,i} + \text{NAV}_{t,i}) - \text{NAV}_{t-1,i}$$

denote the dollar NAV Euler error corresponding to the quarter-end net-cash-flow approximation. Multiplying $\eta_{t,i}^{\$}(\theta)$ by $\Psi_{0,t-1}(\theta)$ and summing would yield a valid moment, and for fully liquidated funds it would telescope. But for the young funds of actual interest it would recreate the long-horizon terminal-NAV term:

$$\sum_{t=1}^{T_i} \Psi_{0,t-1}(\theta) \eta_{t,i}^{\$}(\theta) = \sum_{t=1}^{T_i} \Psi_{0,t}(\theta) CF_{t,i} + \Psi_{0,T_i}(\theta) \text{NAV}_{T,i} - \text{NAV}_{0,i}.$$

For an unseasoned fund, $\text{NAV}_{T,i} \neq 0$, so discounting the quarterly Euler errors back to time zero reconstructs exactly the large terminal-NAV present-value term that generates instability. The normalized construction in Section 5.2 sacrifices path-by-path telescoping in order to preserve the short-horizon dependence of the young-fund block on the candidate SDF.

F Research Agenda

This appendix collects extensions that the framework accommodates but the main paper does not pursue. Each is a well-posed follow-up project; none is required for the buyout application.

F.1 Vintage-stratified factor models

Instead of one β across all vintages, define vintage-bucket-specific loadings β_b for buckets $b \in \{B_1, \dots, B_K\}$ collecting contiguous vintage windows. The parameter vector becomes $\theta = (a, \beta_{B_1}, \dots, \beta_{B_K})'$ and the

blocked framework applies without modification: the held-out loss automatically penalizes overfitting, and the identification requirement becomes enough estimation vintages with linearly independent sensitivities *per bucket's parameters*—a binding constraint when whole blocks of vintages are held out, so bucket models need wider estimation sets or stronger pooling.

Floating buckets. Overlapping windows centered on each vintage year (e.g., a five-year window centered on 2010) give smooth estimates of how exposures evolve without sharp boundary effects; the window width is a tunable design parameter ranked by held-out loss.

Strategy-specific buckets. Buckets defined jointly over vintage year and strategy (“2002 buyout factor,” “2020 venture factor”) let the held-out loss compare pooled versus bucket β , narrow versus wide windows, and strategy-specific versus pooled buckets.

Partial pooling and fused exposure paths. Free buckets get noisy quickly. Two regularizations keep the bucket interpretation while stabilizing the path, in the spirit of regularized GMM for time-varying parameter models [Cui et al., 2024]: hierarchical shrinkage $\beta_b = \bar{\beta} + u_b$ with $\sum_b \|u_b\|_2^2$ penalized, and a fused-lasso path penalty $\lambda \sum_{b \geq 2} \|\beta_b - \beta_{b-1}\|_1$ that encourages neighboring buckets to share an exposure unless the data support a break. The tuning parameter λ is another design choice ranked by held-out loss.

F.2 Age-varying exposures along the fund life

PE risk arguably varies at least as much over a fund’s own life as across cohorts: a young fund is illiquid, partly drawn, and conservatively marked (low measured exposure during the J-curve); near exit it behaves more like levered public equity. Writing $a_{i,t}$ for fund age, the one-period SDF becomes $M_{t-1,t}(\theta)$ with loadings $\beta(a_{i,t})$, a step function over age buckets or a smooth path regularized along the age axis. Identification comes from observing many funds at each age and from the within-fund time series of the Euler block; the held-out loss arbitrates pooled versus vintage versus age versus strategy- $\times$ -age loadings. A caveat: fund age, calendar period, and vintage are linearly dependent (period = vintage + age), the classic age–period–cohort collinearity. The macro “period” dimension is carried by the factor realizations inside the SDF, so age and vintage loadings are separated only through the factor structure and the SDF’s nonlinearity—avoid stacking unrestricted linear vintage and age exposures simultaneously, identify one life-cycle axis at a time, or tie them together with the shrinkage path above.

F.3 Alternative alpha definitions

The design vector can carry alpha definitions beyond the cumulative baseline: (i) the DLP12-style per-period intercept alpha, retained as a comparison and the main consumer of the robust-inference toolkit of Appendix B; (ii) an error-term alpha ($\mathbb{E}[\epsilon]$) as in the machine-learning approach of Tausch and Pietz [2024]. The empirical question is not only which SDF factors perform best, but which alpha definition yields stable and predictive PE pricing moments.

F.4 Moment weighting and block refinements

The baseline uses equal vintage weights and a diagonal quadratic form. Refinements include value weights, shrinkage or block-cluster weighting toward efficient GMM weighting of the vintage vector, and strategy–vintage cells in multi-strategy samples (with the rank condition then counted at the cell level). On the holdout side, the block length d and buffer g can be studied as design parameters in their own right—including data-driven feasibility rules tied to the Jacobian condition number per block—and random (non-contiguous)

vintage subsets provide a robustness check on the contiguity choice at the cost of weaker dependence control.

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